

Learning Disciplinary Domain Distributions for Swedish University Courses: A Multi-Label Study with KB-BERT

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Abstract

University courses in Sweden are assigned disciplinary domains, which determine state funding allocations. We examine automatic classification of Stockholm University courses as a multi-label and label distribution learning task, using course plans and metadata from a 2023 Ladok extract. We compare TF-IDF-based linear models with a fine-tuned KB-BERT model, evaluating both domain set prediction and percentage-based domain distributions. KB-BERT significantly outperforms TF-IDF baselines in multi-label classification, achieving a subset accuracy of 90.8% compared to 82.9% for the strongest baseline (Linear SVC). For distribution prediction, direct training with a KL-divergence loss yields a 48% lower mean absolute error than post-hoc normalization of binary predictions (1.44 vs. 2.79 percentage points). Error analysis, carried out together with a domain expert, indicates that most large discrepancies between predictions and gold labels reflect administrative conventions, program constraints, or historical policies rather than content misunderstanding. The findings suggest that content-based models can support administrators by highlighting courses whose classifications merit review.

1 Introduction

University courses are assigned to disciplinary domains for purposes such as internal planning, national statistics, and state funding allocation. In Sweden, these domains, called *utbildningsområden* (UO), determine how much funding universities receive per student and credit. Misclassification therefore has direct financial consequences and can distort how resources are distributed among faculties.

At Stockholm University (SU), course classifications are decided by administrators and sub-

ject experts following national regulations and local guidelines (Utbildningsdepartementet, 1992; Regeringskansliet, 2024; Stockholms universitet, 2024). This is a complex task since many courses are interdisciplinary by nature and classified by percentage-based distributions (e.g., 70% humanities, 30% natural sciences). From a machine-learning perspective, this constitutes a document classification problem with both multi-label and label-distribution characteristics.

Multi-label classification (MLC) methods include problem-transformation approaches such as one-vs-rest training, as well as algorithm-adaptation methods that extend classical classifiers to handle multiple labels (Herrera et al., 2016). Typically, MLC treats each label as binary, either present or absent, which cannot capture the graded nature of percentage-based allocations. Label Distribution Learning (LDL) addresses this limitation by associating each label with a *degree* describing how strongly it applies to an instance (Geng, 2016). The SU course-classification task, with its explicit percentage allocations, motivates exploring both frameworks.

Transformer-based models have achieved strong performance on document classification tasks (Adhikari et al., 2019), though (Park et al., 2022) showed that complex models do not always outperform simpler baselines.

Machine learning for classifying Swedish educational content has received limited attention. (Andrén and Normark, 2020) used both supervised and unsupervised methods to classify syllabi from Malmö University as approved/not approved, achieving high accuracy on held-out test data but finding that models failed to generalize to new syllabi evaluated by domain experts. Their study highlights the importance of expert validation in educational classification tasks. More recently, (Kann et al., 2025) developed rule-based NLP tools for automatic analysis of learning out-

comes across four Swedish universities, focusing on cognitive complexity levels.

To our knowledge, no prior work has addressed the classification of Swedish university courses according to disciplinary funding categories (*utbildningsområden*), nor the multi-label and distributional aspects inherent in this task.

In this study, we fine-tune KB-BERT, a Swedish transformer model pretrained by the National Library of Sweden (Malmsten et al., 2020), which has previously been successfully adapted for Swedish text classification tasks (Hägglöf, 2023; Fano and Haffenden, 2022).

We compare TF-IDF-based linear models with fine-tuned KB-BERT on SU course descriptions, evaluating both multi-label performance and the ability to predict full percentage-based distributions.

For distribution prediction, we consider two variants: (i) a baseline that first produces independent sigmoid scores for each label and then (post-hoc) converts them into a distribution by normalising the scores to sum to one, and (ii) a model with a softmax output layer trained directly to match gold distributions using KL-divergence loss.

Additionally, we conduct two instances of qualitative error analysis in collaboration with an expert to determine whether prediction errors reflect genuine content misclassification or administrative conventions not captured in course descriptions.

Concretely, we address the following research questions:

1. How accurately can machine learning methods reproduce existing course classifications, and how do TF-IDF-based approaches compare to transformer-based models (KB-BERT)?
2. To what extent can models predict the percentage-based distribution of disciplinary domains, and does direct distributional learning outperform post-hoc normalization of binary predictions?
3. What systematic patterns characterize prediction errors, and what do these patterns reveal about the relationship between course content and administrative practice?

Based on prior work demonstrating the effectiveness of pretrained language models for text

classification (Tarekegn et al., 2024), and the identification of our task as a case of LDL (Geng, 2016), we formulate the following hypotheses:

- H1** Fine-tuned KB-BERT will significantly outperform TF-IDF-based linear classifiers on multi-label classification metrics.
- H2** Direct distributional prediction will yield more accurate percentage allocations than post-hoc normalization of binary predictions.

2 Data

2.1 Corpus

The corpus was provided by SU Administration as a 2023 extract from Ladok, the national student information system used by Swedish universities. It contains syllabi with metadata including titles, descriptions, learning objectives, and disciplinary domain classifications. Decision dates for included course versions span from 2005 to early 2024.

Each course is classified into one or more of $K = 10$ disciplinary domains (“utbildningsområden”, UO), with percentage allocations for each assigned domain. For example, a course in Psychology is assigned 60% Medicine and 40% Social Science.

2.2 Label Distribution

The label distribution is imbalanced. The three largest education areas, Humanities (HU), Social Sciences (SA), and Natural Sciences (NA), appear in approximately 35%, 34%, and 19% of course versions, respectively. Less frequent domains are: Other (ÖV) accounts for 8%, while Teacher Education (LU), Medicine (ME), Law (JU), Technology (TE), and Health Care (VÅ) each occur in roughly 4–6%. Widening Participation (VU) is the rarest label at 1.9%.

81.8% of the courses have one UO label, 16.6% two labels. Three labels are the rarest (1.6% or 157 courses). In total, there are 38 distinct label combinations. Multi-label assignments are dominated by Social Sciences pairings: SA appears in 65% of multi-label combinations (e.g., SA+TE, SA+ME).

2.3 Preprocessing

The raw corpus contains 12,949 rows in long format (one row per UO label assignment). A *course version* is defined as a unique (course ID, decision date) pair, since both content and classifications

may change over time. Aggregating UO codes and percentages by course version leaves us with 9,770 instances, representing 4,880 unique courses.

Text fields (title, description, learning objectives) were concatenated and normalized: removing markup, collapsing whitespace, and stripping special characters.

2.4 Train–Validation Split

We perform a stratified split at the course level (by primary UO), then propagate assignments to all corresponding versions. This yields 7,865 training instances (3,904 courses) and 1,905 validation instances (976 courses).

3 Methodology

Here we describe the experimental setup used to address the research questions. For RQ1 and RQ2, we frame the task in two complementary ways: as MLC (predicting which UO codes apply) and as distribution prediction (estimating percentage allocations). For RQ3, we conduct two forms of error analysis together with an expert.

3.1 Problem Formulation

Given a course description text x , the goal is to predict an assignment to one or more of $K = 10$ disciplinary domains. The corpus provides two forms of supervision: (1) a binary vector $\mathbf{y} \in \{0, 1\}^K$ indicating which domains are assigned, and (2) a probability vector $\mathbf{p} \in [0, 1]^K$ where $\sum_k p_k = 1$, derived from percentage allocations.

3.2 Baseline Models

We employ TF–IDF features with character n-grams (length 3–5, word boundaries) combined with one-vs-rest linear classifiers: **Logistic Regression** with balanced class weights, and **Linear SVC**. Both produce binary predictions for each UO code.

3.3 Transformer Models

We fine-tune KB-BERT (Malmsten et al., 2020), a Swedish BERT model from the National Library of Sweden. Two training objectives are explored:

Binary Multi-Label Classification. A classification head with sigmoid activation is trained using binary cross-entropy loss. Each domain is treated as an independent binary classification

task. For distribution prediction, the sigmoid outputs (not constrained to sum to 1) are post-hoc normalized to sum to 1.

Direct Distribution Prediction. A softmax output layer trained with KL divergence loss directly models the percentage allocation as a probability distribution:

$$\mathcal{L} = D_{\text{KL}}(\mathbf{p} \parallel \hat{\mathbf{p}}) = \sum_k p_k \log \frac{p_k}{\hat{p}_k} \quad (1)$$

This Label Distribution Learning formulation directly optimizes for percentage allocations.

3.4 Training Details

Transformer models are fine-tuned for 3 epochs using AdamW (learning rate 2×10^{-5} , weight decay 0.01), batch size 8, and a maximum sequence length of 512 tokens.

3.5 Evaluation

Multi-Label Metrics. We report:

- **subset accuracy** (exact match of all labels)
- **micro-F1** (global F1 across all assignments)
- **macro-F1** (unweighted average per-label F1)

Distribution Metrics. We report:

- **mean absolute error** (MAE) in percentage points
- **top-1 accuracy** (primary domain match),

3.6 Statistical Testing

To assess statistical significance, we employ paired bootstrap resampling (Efron and Tibshirani, 1994) with $B=10,000$, forming 95% percentile confidence intervals. Differences are considered significant at $\alpha=0.05$

3.7 Error Analysis

To investigate systematic patterns (RQ3), we employ a two-stage analysis involving expert consultation from SU Administration.

Data Exploration. We use **LDA topic modeling** (Blei et al., 2001) ($K = 10$) to identify courses where content-based topic assignments diverged from administrative UO labels. Each topic is mapped to its most frequent UO label, creating a topic-to-UO correspondence. Courses are then ranked by divergence between their LDA topic distribution and their assigned UO label(s). Approximately 50 high-divergence cases were reviewed with the SU domain expert, focusing on

Table 1: Model comparison, multi-label classification (RQ1). All BERT-SVC differences significant at $p < 0.001$.

Metric	LogReg	SVC	BERT
Subset Acc.	77.8	82.9	90.8
Micro-F1	87.8	90.0	93.6
Macro-F1	83.6	85.8	90.5

courses where the topic model suggested a different primary UO than the gold label. We also computed topic entropy $H = -\sum_k \theta_k \log \theta_k$ for each course to assess whether multi-label assignments reflect greater content diversity than single-label courses.

Model Error Analysis. We select 25 cases for expert review using three criteria: (1) *high-MAE* predictions (>10 percentage points error), representing the model’s largest mistakes; (2) *high-uncertainty* predictions, where the model’s output entropy indicated low confidence; (3) *primary-mismatch* cases where the predicted primary UO differed from gold despite low overall MAE. For each case, the expert is asked to examine gold labels, model predictions, and provide context on administrative classification conventions.

4 Results

4.1 Multi-label Classification (RQ1)

KB-BERT achieves subset accuracy of 90.8%, a 7.9 percentage point improvement over Linear SVC (82.9%; 95% CI [6.30, 9.34], $p < 0.001$). Micro-F1 improves by 3.6 points and macro-F1 by 4.7 points.

4.2 Distribution Prediction (RQ2)

Table 2 compares BERT training objectives for percentage prediction. The distributional model substantially outperforms binary classification with post-hoc normalization.

Top-1 accuracy improves from 85.25% to 92.39% ($\Delta = +7.14$ pp, $p < 0.001$). MAE drops from 2.79 to 1.44 percentage points, a 48% relative improvement.

Per-Label Performance. Table 3 shows MAE by UO category. Social Sciences (SA) exhibits the highest error (4.38 pp), possibly because it participates in 65% of multi-label combinations and shares conceptual overlap with HU, LU, and ME. The three largest categories (SA, NA, HU) account

Table 2: Distribution prediction results (RQ2). All differences significant at $p < 0.001$.

Metric	Binary	Distrib.
Top-1 Acc.	85.3	92.4
MAE (pp) ↓	2.79	1.44

Table 3: Per-label MAE (percentage points) for the distributional model. Larger, more ambiguous domains show higher error.

Code	Domain	MAE (pp)
SA	Social Sciences	4.38
NA	Natural Sciences	2.23
HU	Humanities	2.19
LU	Teacher Education	1.99
ÖV	Other	1.40
ME	Medicine	0.80
VÅ	Health Care	0.69
TE	Technology	0.58
JU	Law	0.46
VU	Widening Participation	0.19

for most prediction error. Other domains, such as Law (JU) and Technology (TE), show substantially lower MAE (<0.6 pp).

4.3 Error Analysis (RQ3)

Data Exploration. LDA topic modeling identified courses where unsupervised topic assignments diverged from administrative labels. Expert consultation revealed that this divergence largely reflects administrative conventions in the gold standard. Three patterns emerged: (1) regulatory stipulations assigning ÖV to certain language courses for funding purposes (Regeringskansliet, 2024); (2) program restrictions preventing LU classification for didactics courses outside the teacher program; (3) historical policy fixing linguistics courses at 50/50 HU/NA due to a departmental merger. Several flagged courses have since been administratively reclassified, suggesting topic modeling can identify review candidates.

Model Errors Analysis of high-MAE BERT predictions confirmed that most large disagreements correspond to administrative patterns rather than content misunderstanding. The model predicts content-appropriate labels (e.g., HU for language courses) that conflict with administratively-determined gold labels. Genuine failures were rare. One example of such a case is “Mental

innehåll” (Philosophy of Mind) was predicted as ME/SA instead of HU.

Topic Entropy No significant difference emerged between single-label ($H = 1.56$) and multi-label ($H = 1.54$) courses, suggesting multi-label assignments reflect administrative partitioning rather than content diversity.

5 Discussion

5.1 Interpretation of Results

Both hypotheses were confirmed with strong statistical significance. The 7.9 percentage point improvement in subset accuracy (KB-BERT vs. Linear SVC) suggests that contextual embeddings seem to capture semantic distinctions that bag-of-words representations miss, confirming recent trends in NLP (Tarekegn et al., 2024).

The advantage of direct distributional learning over post-hoc normalization (H2) supports the theoretical motivation of Label Distribution Learning (Geng, 2016). Binary classification treats labels independently and cannot learn trade-offs between competing categories; the softmax formulation forces the model to allocate probability mass, better matching the structure of percentage-based assignments. The 48% reduction in MAE demonstrates that this architectural choice has a substantial impact.

This finding is reinforced by the error analysis: the model often predicted content-appropriate labels even when they conflict with administratively-determined gold labels. For example, language courses assigned to ÖV for regulatory funding reasons are predicted as HU based on their humanities content. This pattern is consistent with the model having learned content-based representations, though other explanations cannot be ruled out.

5.2 Implications for Practice

Error analysis confirmed that many prediction errors reflect administrative conventions rather than content misunderstanding. This implies that the model may be more useful for *content-based classification review* than for replicating existing administrative assignments. Courses where model predictions diverge strongly from gold labels are candidates for human review, as some may reflect outdated classifications. Indeed, several courses flagged by the LDA analysis have now been reclassified.

5.3 Limitations

Several limitations apply. First, the corpus includes only courses from Stockholm University, so generalization to other contexts is untested. Second, data from 2023 was used due to technical issues with accessing current Ladok records. Third, potential overlap exists between course descriptions and KB-BERT’s pretraining corpus (which includes official government documents), risking inflated performance (Malmsten et al., 2020). Fourth, a single train-validation split was used, limiting variance assessment. Fifth, only 18.2% of courses were multi-label, so distributional predictions focus on edge cases. The error analysis was qualitative and limited in scope; while expert review suggested many “errors” reflect administrative conventions, the relative influence of content versus policy was not quantified. Future work could evaluate generalization to other Swedish universities using the corpus developed by (Björn et al., 2024), or compare advanced MLC architectures (e.g., Tarekegn et al. (2024)) to our softmax baseline.

6 Ethical Considerations

The corpus contains administrative metadata and public course descriptions, but no personal data. Course information is publicly available¹. Automated classification of courses into funding-relevant domains could, in principle, be misused to optimize allocations rather than accurately reflect content. Here, it is worth noting that we view our models as decision-support tools to assist human reviewers, not as replacements for expert judgment.

7 Conclusion

We addressed the task of automatically classifying Swedish university courses into disciplinary domains, framing it as both multi-label classification and label distribution learning. Fine-tuned KB-BERT significantly outperformed TF-IDF baselines on multi-label metrics ($p < 0.001$), and direct distributional training yielded 48% lower MAE than post-hoc normalization of binary predictions.

Error analysis, conducted through LDA topic modeling and expert consultation, revealed that most prediction-gold disagreements reflect admin-

¹ www.su.se/utbildning/utbildningskatalog

istrative conventions, regulatory funding rules, program restrictions, and historical policy, rather than failures of content understanding. This suggests that content-based models may be valuable for identifying courses whose classifications merit review.

The Label Distribution Learning framework is well-suited for tasks where labels have associated proportions. Future work should evaluate generalization to other institutions and comparison to more advanced models.

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Code and Data Availability

All code used for this study is available at www.github.com/fboglind/lis070-su-admin-project. Information about data access is provided in the repository.

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A Evaluation Metrics

A.1 Multi-Label Classification Metrics

Let $\mathbf{y} \in \{0, 1\}^K$ be the gold label vector and $\hat{\mathbf{y}} \in \{0, 1\}^K$ the predicted label vector for K labels across N instances.

Subset Accuracy. The proportion of instances where all predicted labels exactly match all gold labels:

$$\text{Subset Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\mathbf{y}_i = \hat{\mathbf{y}}_i] \quad (2)$$

Micro-F1. F1-score computed globally by aggregating true positives, false positives, and false negatives across all labels:

$$\text{Micro-F1} = \frac{2 \cdot \text{TP}}{2 \text{ TP} + \text{FP} + \text{FN}} \quad (3)$$

Macro-F1. Unweighted average of per-label F1 scores:

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^K \text{F1}_k \quad (4)$$

Hamming Loss. The fraction of labels incorrectly predicted:

$$\text{Hamming} = \frac{1}{NK} \sum_{i=1}^N \sum_{k=1}^K \mathbf{1}[y_{ik} \neq \hat{y}_{ik}] \quad (5)$$

A.2 Distribution Prediction Metrics

Let $\mathbf{p} \in [0, 1]^K$ be the gold distribution and $\hat{\mathbf{p}} \in [0, 1]^K$ the predicted distribution, both summing to 1.

Mean Absolute Error (MAE). Average absolute difference between predicted and gold percentages, reported in percentage points:

$$\text{MAE} = \frac{100}{NK} \sum_{i=1}^N \sum_{k=1}^K |p_{ik} - \hat{p}_{ik}| \quad (6)$$

Top-1 Accuracy. Whether the predicted primary domain matches the gold primary domain:

$$\text{Top-1 Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\arg \max_k p_{ik} = \arg \max_k \hat{p}_{ik}] \quad (7)$$